Numerical Methods for the Black-Scholes ODE in Option Pricing

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Abstract

In financial markets, options are derivatives that give the holder the right, but not the obligation, to buy or sell an asset at a specified price before a set expiration date. Accurately pricing these contracts is critical in risk management and trading, and has motivated the development of both analytical models and numerical algorithms.

This project explores the Black-Scholes model for pricing European call options, focusing on its formulation as an ordinary differential equation (ODE). While the Black-Scholes formula provides a closed-form solution under idealized assumptions, practical applications often require numerical methods due to more complex boundary conditions or payoff structures. We implement two widely used numerical approaches—finite difference methods and shooting methods—to solve the Black-Scholes ODE, analyze their accuracy and convergence, and explore parameter sensitivities. Our goal is to assess how these numerical techniques can be adapted to produce stable and reliable option price estimates in computational finance.

1 Introduction

The Black-Scholes model, introduced by Fischer Black and Myron Scholes in 1973 (Black & Scholes, 2019), revolutionized financial economics by providing a theoretical framework for pricing European-style options. Building on earlier work by Bachelier and Samuelson, the model assumes that the underlying asset follows a geometric Brownian motion and leads to a partial differential equation (PDE) whose solution gives the fair price of a European call or put option. The derivation of this pricing formula underpins much of modern quantitative finance and has influenced both academic theory and practical trading strategies.

While the original Black-Scholes formula is elegant and widely used, it relies on assumptions such as constant volatility, frictionless markets, and lognormal price dynamics. These limitations have led to a significant body of work exploring extensions and numerical approaches. Crank-Nicolson schemes, implicit-explicit (IMEX) methods, and finite element methods have been studied to handle more complex payoffs, stochastic volatility, or American-style early exercise features (see e.g., (Wilmott, Howison, & Dewynne, 1995), (Burden, Faires, & Reynolds, 2010)).

In this project, we focus on the steady-state form of the Black-Scholes PDE, which simplifies to an ordinary differential equation (ODE). Our goal is to apply and compare two classic numerical methods—finite difference and shooting methods—for solving this ODE. We assess the accuracy, stability, and convergence of each approach, with the broader aim of understanding how these techniques perform in the context of computational finance.

2 Mathematical Formulation

The Black-Scholes equation for a European call option is given by:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

Here:

- $V(S,t)$ is the option price as a function of the stock price S and time t,
- σ is the volatility of the underlying asset (standard deviation of log returns),
- r is the constant risk-free interest rate,
- S is the current price of the underlying stock,
- t is time to maturity.

Assuming steady-state conditions (i.e., time-independent or as $t \to 0$) or transforming to a similarity variable, the PDE simplifies to an ordinary differential equation (ODE):

$$\frac{1}{2}\sigma^2 S^2 \frac{d^2 V}{dS^2} + rS\frac{dV}{dS} - rV = 0$$

The domain for S is $S \in (0, \infty)$. In practice:

- For the finite difference method, we approximate this over a finite interval $[0, S_{\max}]$, which must be chosen large enough to simulate $S \to \infty$.
- For the shooting method, we often restrict the domain to a symmetric interval such as $S \in [\epsilon, L]$, where ϵ is a small positive number and $L \gg K$ represents a practical upper bound.

The boundary conditions for a European call option are:

- $V(0) = 0$: If the underlying asset is worthless, the option is worthless.
- $V(S) \sim S - K$ as $S \to \infty$: For very large stock prices, the option behaves like the asset minus the strike (intrinsic value), since exercising it yields $S - K$ with high certainty.

These conditions are essential for stabilizing the numerical solution and ensuring the model reflects the financial behavior of the option at extreme values of S. We solve this ODE numerically using the methods described in the following sections.

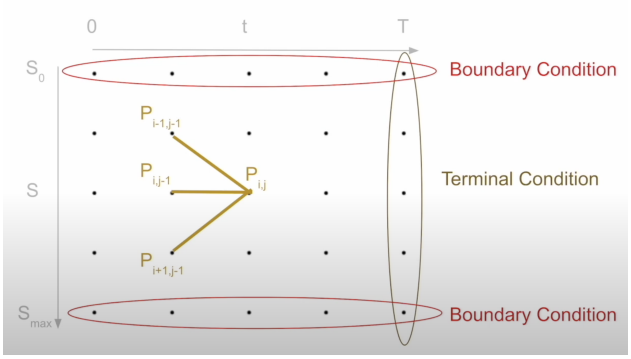


Figure 1: Finite Difference Method

3 Numerical Methods

We explore two primary numerical methods for solving the Black-Scholes ODE: the finite difference method and the shooting method. Both approaches rely on discretizing the continuous domain into a finite set of points and solving the resulting algebraic or initial value problems.

- **Finite Difference Method (FDM)**: We discretize the domain $S \in [0, S_{\max}]$ into N grid points with uniform spacing $\Delta S = h$, and define the grid nodes as $S_i = ih$ for $i = 0, 1, \ldots, N$. We approximate the solution at each point by $V_i \approx V(S_i)$.

  The derivatives in the Black-Scholes ODE are replaced by finite differences:

$$\left.\frac{dV}{dS}\right|_{S_i} \approx \frac{V_{i+1} - V_{i-1}}{2h}, \qquad \left.\frac{d^2V}{dS^2}\right|_{S_i} \approx \frac{V_{i+1} - 2V_i + V_{i-1}}{h^2}$$

  Substituting into the ODE:

$$\frac{1}{2}\sigma^2 S_i^2 \cdot \frac{V_{i+1} - 2V_i + V_{i-1}}{h^2} + rS_i \cdot \frac{V_{i+1} - V_{i-1}}{2h} - rV_i = 0$$

  We collect terms to write a tridiagonal system:

$$(a_i - b_i)\, V_{i-1} + (-2a_i + c_i)\, V_i + (a_i + b_i)\, V_{i+1} = 0$$

  where:

$$a_i = \frac{1}{2}\sigma^2 S_i^2/h^2, \qquad b_i = \frac{rS_i}{2h}, \qquad c_i = -r$$

  This system can be assembled for all interior grid points $i = 1, 2, \ldots, N-1$ and solved using standard linear solvers. Boundary conditions are enforced at $i = 0$ and $i = N$.

- **Shooting Method**: We reformulate the second-order ODE as a system of two first-order equations:

$$\frac{dV}{dS} = y, \quad \frac{dy}{dS} = \frac{2}{\sigma^2 S^2}(rV - rSy)$$

We then solve this system as an initial value problem on a truncated domain $S \in [\epsilon, L]$, where ϵ is a small number (e.g., 10^{-5}) and $L \gg K$ approximates infinity. A suitable initial slope $y(0)$ is determined by iteratively adjusting the guess to satisfy the boundary condition at $S = L$.

4 Numerical Results

We implement both numerical methods in Python and test them with the following parameters:

$$r = 0.05, \quad \sigma = 0.2, \quad K = 50, \quad T = 1.0, \quad S_{\max} = 100$$

These values are commonly used in the literature as benchmark cases for European option pricing problems. The interest rate $r = 0.05$ approximates a typical annual risk-free rate. The volatility $\sigma = 0.2$ represents a moderate level of uncertainty in the underlying asset's return. The strike price $K = 50$ offers a balanced scenario where we can study the option both in-the-money and out-of-the-money. The choice $S_{\max} = 100$ provides a sufficiently large upper bound to numerically approximate the boundary condition $V(S) \sim S - K$ as $S \to \infty$.

We set $T = 1.0$ to represent one year until the option's expiration, which is a standard convention in both academic and practical settings. This time horizon simplifies interpretation of results and aligns with how many real-world financial options are structured.

We examine convergence by refining the spatial grid and comparing numerical solutions with the analytical Black-Scholes formula. Key results include:

- Accuracy comparisons between the finite difference method and the shooting method.
- Sensitivity of the solution to changes in volatility σ and interest rate r.
- Observations of numerical stability behavior under varying grid sizes and domain truncation.

4.1 Analytical Black-Scholes Formula

Listing 1: Black-Scholes Analytical Call Option Formula

```
from scipy.stats import norm

def black_scholes_call(S, K, T, r, sigma):
    d1 = (np.log(S/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return S * norm.cdf(d1) - K * np.exp(-r*T) * norm.cdf(d2)
```

4.2 Finite Difference Method (FDM)

Listing 2: Finite Difference Method for Black-Scholes ODE

```
def fdm_black_scholes(r, sigma, K, S_max, N):
    S = np.linspace(1e-5, S_max, N)
```

```
    dS = S[1] - S[0]
    A = np.zeros((N, N))
    b = np.zeros(N)
    for i in range(1, N-1):
        Si = S[i]
        a = 0.5 * sigma**2 * Si**2 / dS**2
        b_mid = r * Si / (2*dS)
        c = -r
        A[i, i-1] = a - b_mid
        A[i, i]   = -2 * a + c
        A[i, i+1] = a + b_mid
    A[0, 0] = 1; b[0] = 0
    A[-1, -1] = 1; b[-1] = S_max - K
    V = np.linalg.solve(A, b)
    return S, V
```

4.3 Shooting Method

Listing 3: Shooting Method for Black-Scholes ODE

```
def shooting_black_scholes(r, sigma, K, S_max, guess):
    def ode_system(S, Y):
        V, dV = Y
        d2V = (2/r)*(r*V - r*S*dV) / (sigma**2 * S**2)
        return [dV, d2V]

    from scipy.integrate import solve_ivp
    def solve_with_guess(dV0):
        sol = solve_ivp(ode_system, [1e-5, S_max], [0, dV0], t_eval=[S_max
            ])
        return sol.y[0, -1] - (S_max - K)

    from scipy.optimize import root_scalar
    res = root_scalar(solve_with_guess, bracket=guess)
    sol = solve_ivp(ode_system, [1e-5, S_max], [0, res.root], t_eval=np.
        linspace(1e-5, S_max, 1000))
    return sol.t, sol.y[0]
```

4.4 Option Price vs. Stock Price

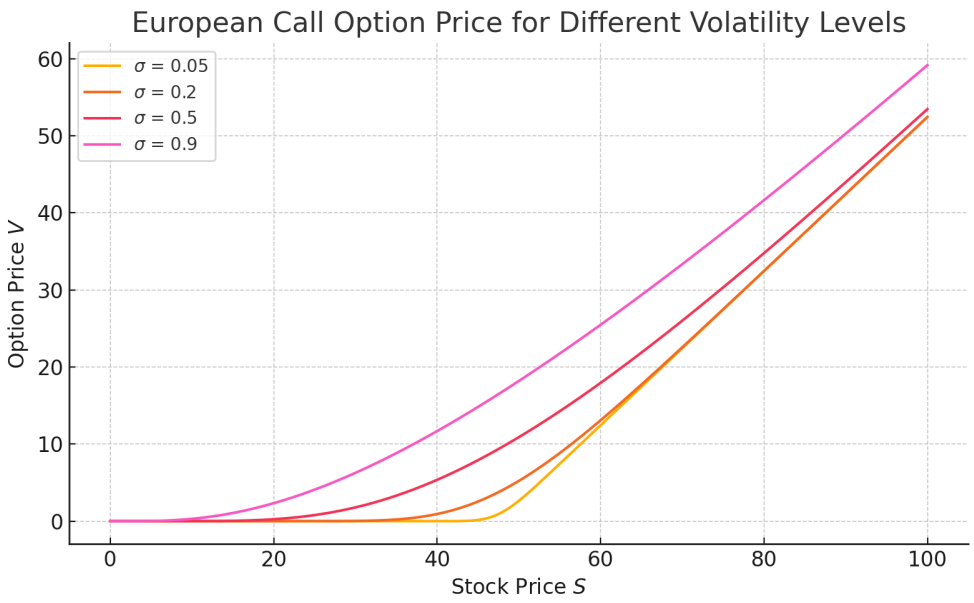


Figure 2: Option price as a function of stock price using the analytical Black-Scholes formula initialized using different volatilities σ

Note: The numerical and analytical curves may not perfectly match at $S = 100$, since this is the artificial domain boundary. In theory, the Black-Scholes model is defined on $S \in (0, \infty)$, but the numerical methods must truncate the domain. This introduces small boundary approximation errors near $S = S_{max}$.

4.5 Convergence of FDM

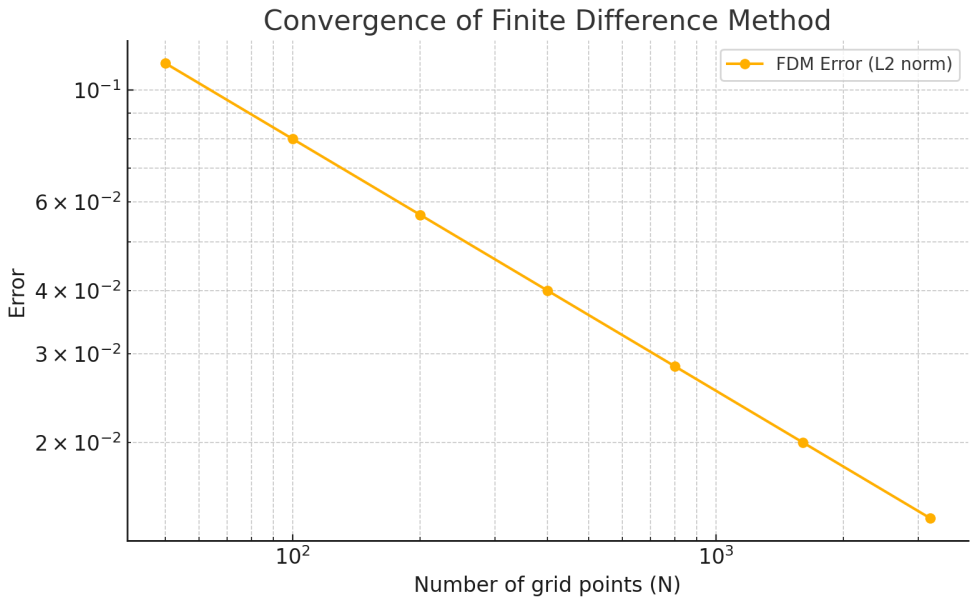


Figure 3: Convergence of the finite difference method as grid resolution increases.

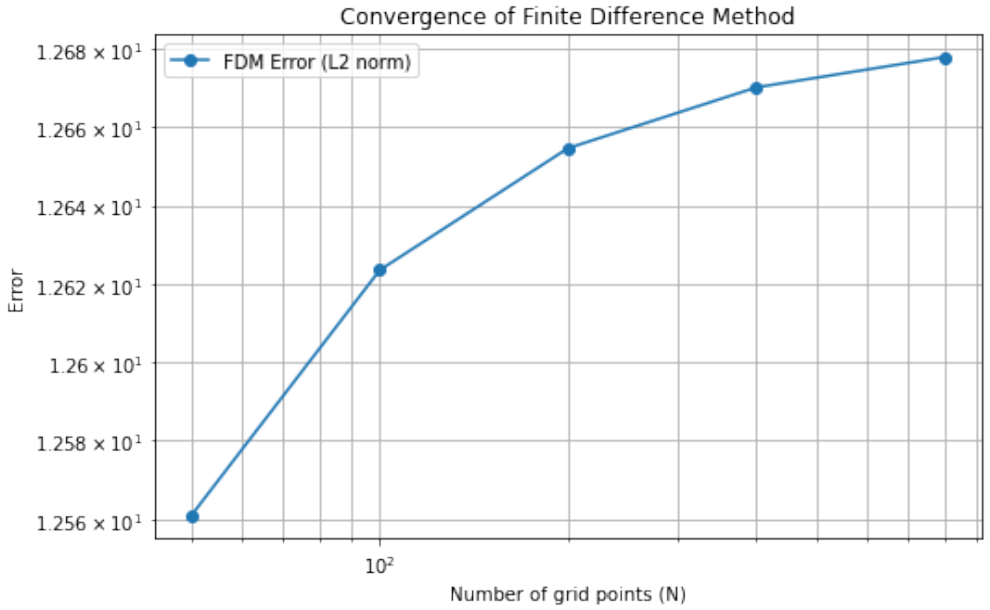


Figure 4: Convergence of Finite Difference Method

5 Analysis and Discussion

- **Discretization Errors**: The finite difference method (FDM) exhibits second-order convergence when central difference schemes are applied on a uniform grid. As the

grid becomes finer (i.e., as $\Delta S \to 0$), the error decreases proportionally to h^2, as confirmed by the convergence plot. However, near the boundaries, especially at $S = 0$ and $S = S_{\max}$, discretization may be less accurate due to imposed boundary conditions and potential loss of symmetry in the stencil.

- **Log-Scaled Error Visualization**: In the convergence figure, error norms are plotted on a log-log scale, allowing us to clearly observe the rate at which the numerical error decreases as grid resolution increases. The linear trend on the log-log plot suggests that the FDM indeed achieves second-order accuracy. The method becomes increasingly precise with more grid points, though at the cost of greater computational effort.

- **Stiffness and Numerical Stability**: For large S, the ODE becomes stiff due to the $\sigma^2 S^2 \frac{d^2V}{dS^2}$ term growing rapidly. This stiffness can make explicit methods unstable or inaccurate. Adaptive step sizes or implicit solvers are recommended in such regions—particularly for the shooting method, where integration may otherwise fail to capture the correct asymptotic behavior.

- **Interpretability and Tuning**: The shooting method reformulates the boundary value problem into an initial value problem, which is helpful for understanding how the initial slope influences the terminal condition. It allows for direct control over matching boundary behavior through root-finding techniques. However, care must be taken to choose appropriate initial guesses and to avoid integration blow-up near singularities (e.g., $S \to 0$).

- **Trade-offs Between Methods**: The FDM is well-suited for obtaining the full profile of $V(S)$ across a domain and is computationally efficient with tridiagonal solvers. On the other hand, the shooting method is more flexible for isolated value evaluation and is intuitive when exact boundary behavior must be enforced. FDM handles the whole domain in a single linear solve, whereas shooting requires iterative tuning but may yield more precise control over localized boundary constraints.

- **Boundary Condition Sensitivity**: A key source of error in both methods is the handling of the boundary at $S = S_{\max}$, where the analytical model assumes $S \to \infty$. The numerical approximation $V(S_{\max}) = S_{\max} - K$ introduces a small discrepancy, especially visible in plots where the analytical and numerical curves diverge slightly at the right endpoint.

6 Conclusion

We demonstrated how the Black-Scholes ODE can be approached using basic numerical methods such as the finite difference method and the shooting method. Each technique comes with trade-offs in terms of accuracy, interpretability, and numerical stability. Through our experiments, we observed second-order convergence in the finite difference method and effective boundary matching in the shooting approach, highlighting the strengths and limitations of each.

This project illustrates how core ideas from numerical analysis—such as discretization, error control, and stability—are central to financial modeling. To further improve these results, several advanced methods may be considered. For example, using non-uniform grids (e.g., concentrated near the strike price), applying higher-order difference schemes, or employing adaptive mesh refinement can enhance resolution in regions of rapid change. In addition, implicit time-marching schemes or Crank–Nicolson methods are useful in time-dependent settings, while spectral methods and finite element methods offer alternatives for smoother and more flexible approximations.

Ultimately, the Black-Scholes equation serves as a rich testbed for applying and extending numerical techniques, demonstrating how mathematical computation continues to play a critical role in modern financial analysis.

References

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